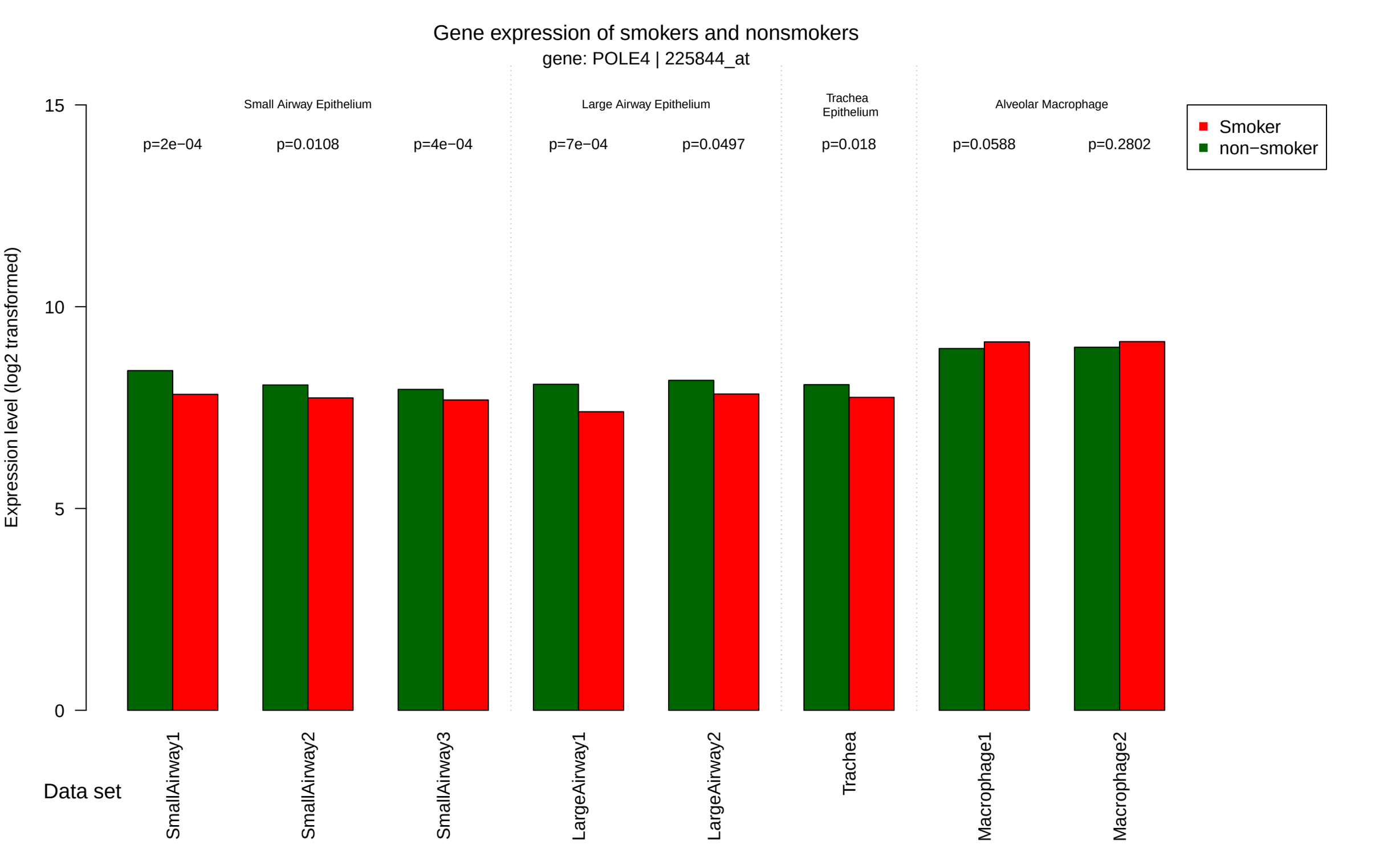
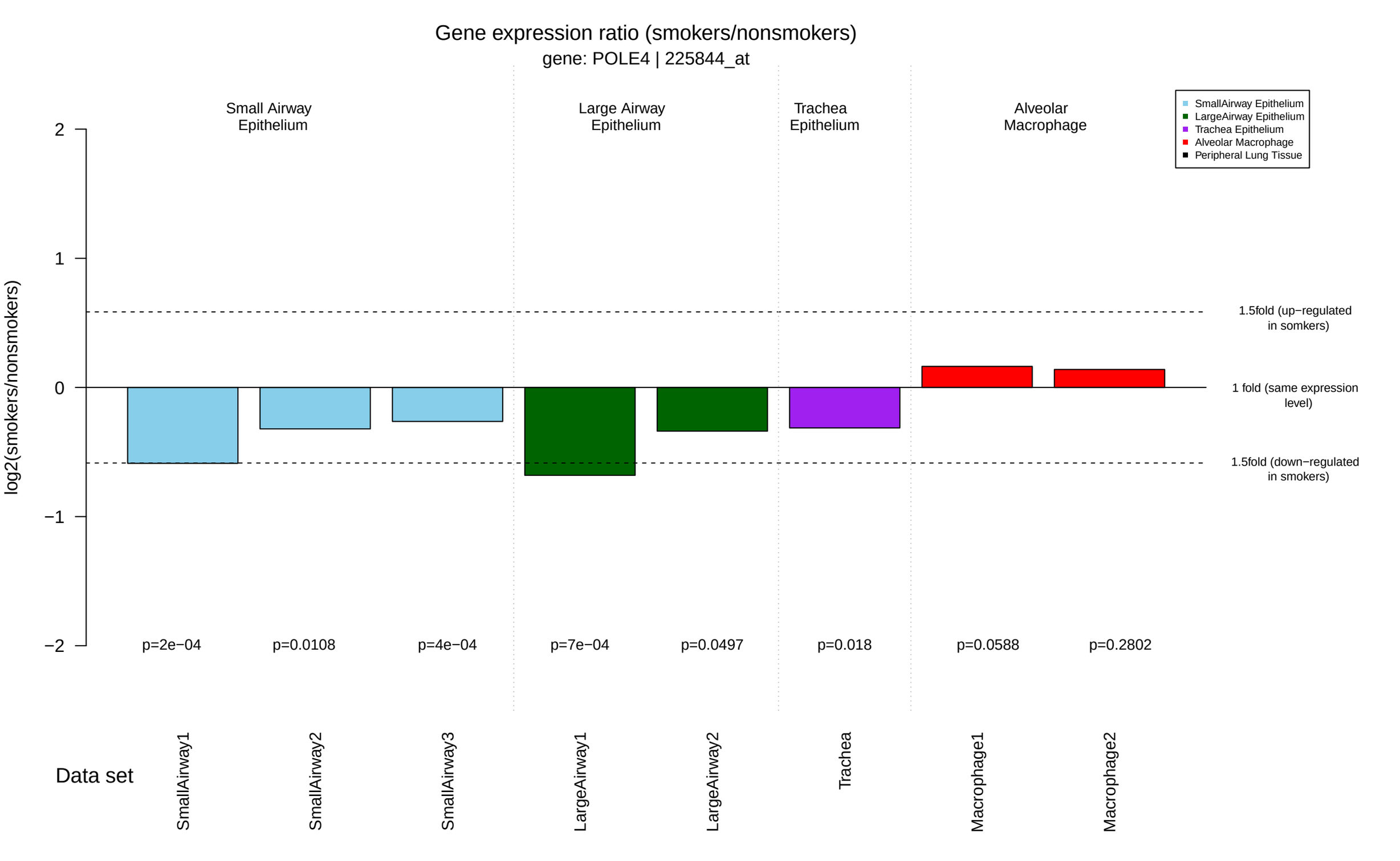
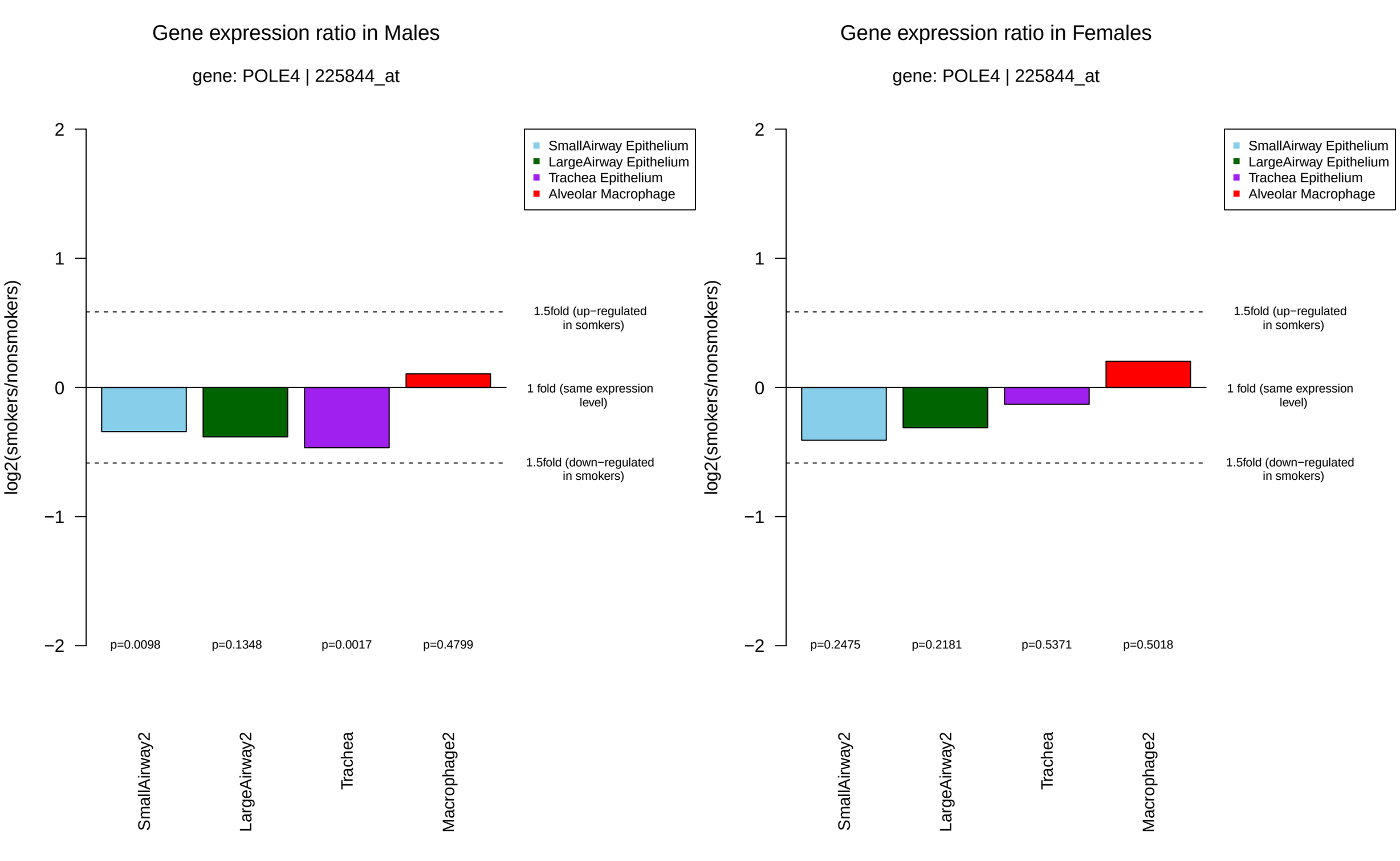
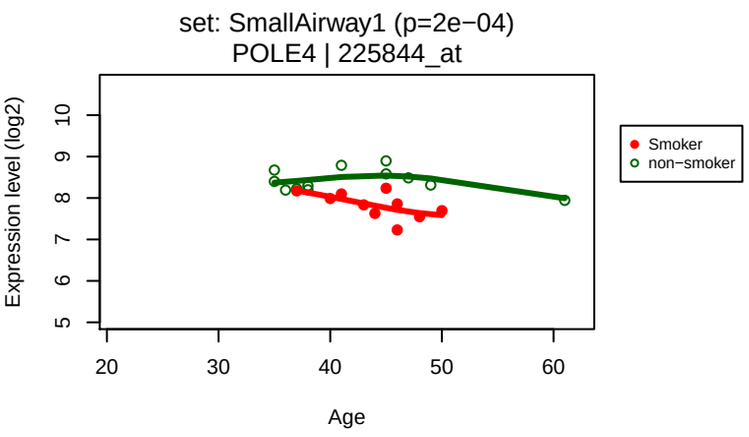
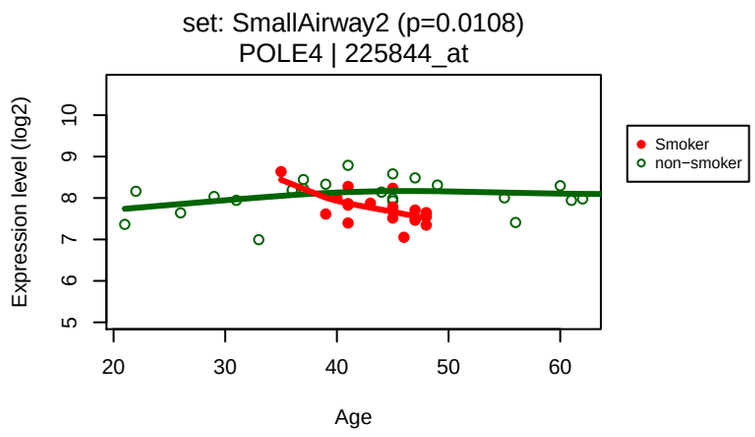
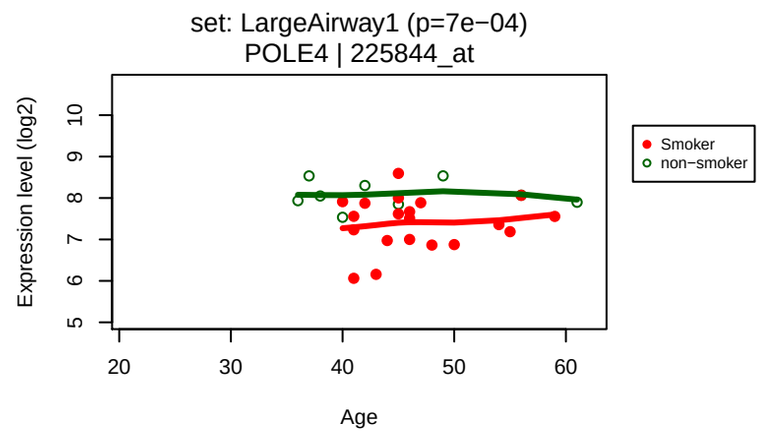

Gene expression of smokers and nonsmokers
gene: POLE4 | 225844_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=2e−04
p=0.0108
p=4e−04
p=7e−04
p=0.0497
p=0.018
p=0.0588
p=0.2802
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

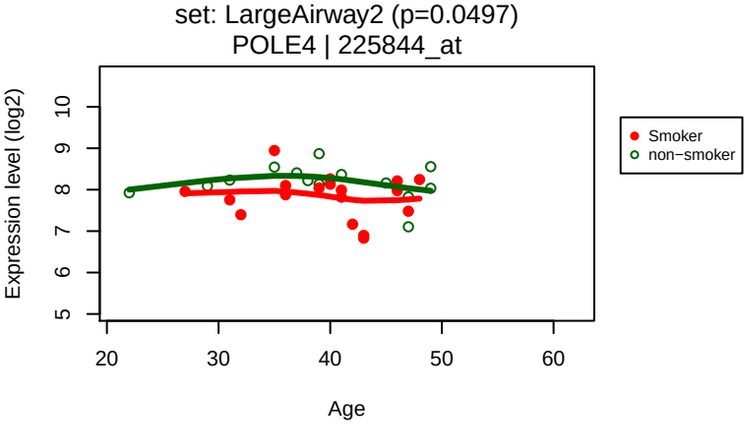

Gene expression ratio (smokers/nonsmokers)
gene: POLE4 | 225844_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=2e−04
p=0.0108
p=4e−04
p=7e−04
p=0.0497
p=0.018
p=0.0588
p=0.2802
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

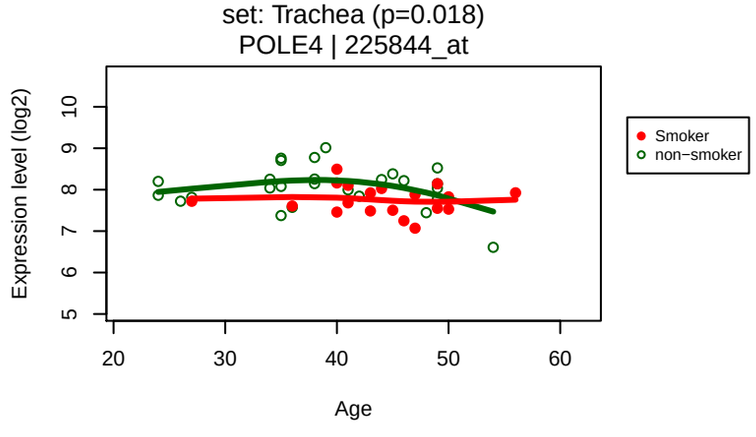

Gene expression ratio in Males
gene: POLE4 | 225844_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0098
p=0.1348
p=0.0017
p=0.4799
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLE4 | 225844_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2475
p=0.2181
p=0.5371
p=0.5018
SmallAirway2
LargeAirway2
Trachea
Macrophage2

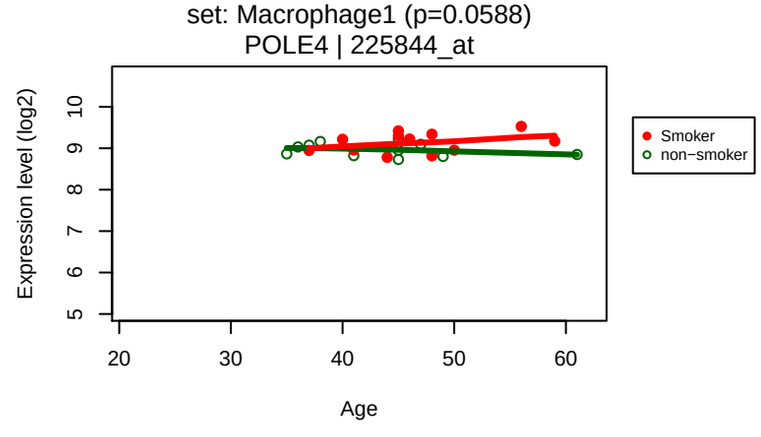

set: SmallAirway1 (p=2e−04)
POLE4 | 225844_at
Expression level (log2)
Age
Smoker
non−smoker

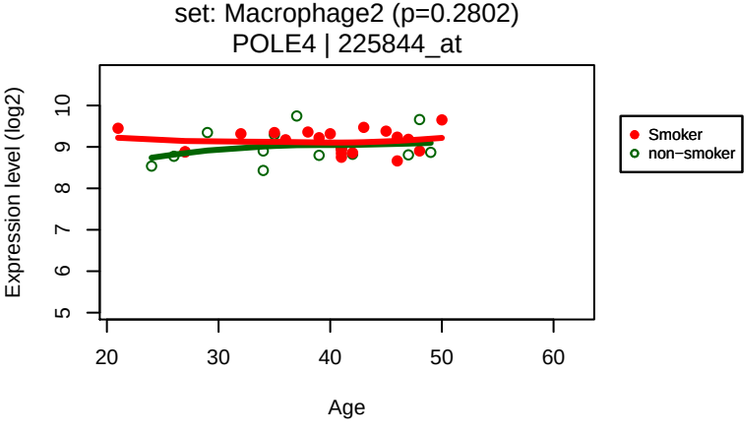

set: SmallAirway2 (p=0.0108)
POLE4 | 225844_at
Expression level (log2)
Age
Smoker
non−smoker

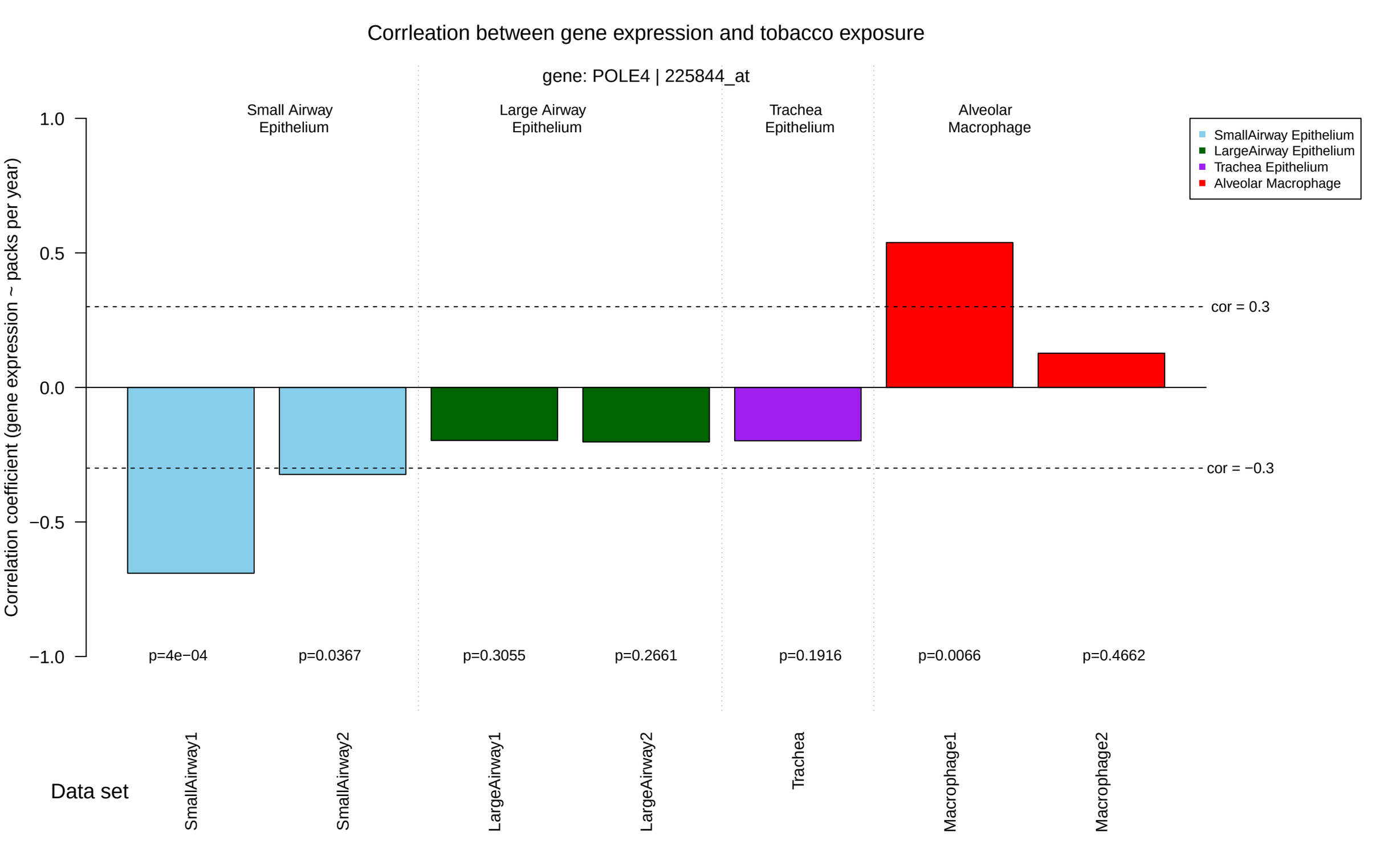

set: LargeAirway1 (p=7e−04)
POLE4 | 225844_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0497)
POLE4 | 225844_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.018)
POLE4 | 225844_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0588)
POLE4 | 225844_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2802)
POLE4 | 225844_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLE4 | 225844_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=4e−04
p=0.0367
p=0.3055
p=0.2661
p=0.1916
p=0.0066
p=0.4662
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2